

Lunch Stealing

CITS3001 problem **lunchstealing** - Time limit: 3s

It is a nice afternoon at the Derbarl Yerrigan, and the black swans have decided that today is a feast day. Students are sprawled along the river with their lunches, and the flock means to make off with as many as it can.

A swan will snatch a student's lunch unless that particular student intimidates it, in which case it keeps a wary distance and leaves that lunch alone. Swans are only able to carry one lunch in their beaks and cannot take the same lunch.

Input

The first line contains two integers S and L ($1 \leq S, L \leq 100$): the number of swans and the number of student lunches. Swans are numbered 1 to S and lunches 1 to L .

Each of the next S lines describes one swan, in order from swan 1 to swan S . The line begins with the swan's own number, followed by the (distinct) numbers of the students it is too scared to steal from. A line containing only the swan's number means that swan fears no one and could take any lunch.

Output

Output a single integer: the most lunches the swans can steal, given that each swan takes at most one lunch and each lunch is taken by at most one swan.

Sample Tests

Sample Input 1

3 3
1 3
2 1
3 2

Sample Output 1

3

Swan 1 fears student 3 (so it may take lunch 1 or 2), swan 2 fears student 1, and swan 3 fears student 2. Assigning swan 1 to lunch 2, swan 2 to lunch 3 and swan 3 to lunch 1 lets all three swans eat.

Sample Input 2

```
3 3
1 3
2 2 3
3 1
```

Sample Output 2

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3
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Sample Input 3

```
3 3
1 2 3
2 2 3
3 2 3
```

Sample Output 3

```
1
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All three swans are scared of students 2 and 3, so every swan can only approach student 1. Just one of them gets a lunch.

Note

A swan that fears every student, or a lunch that no swan dares approach, is possible.